

Performance Evaluation of VGG16 Classifier for Tiny PCB Defect Detection

Research Project: Semantic Aware Tiny Defect Detection in PCB using CNN

May 3, 2026

Abstract

This report presents the classification results of a VGG16-based convolutional neural network trained to detect six types of tiny PCB defects: *open circuit*, *short circuit*, *mousebite*, *spur*, *copper*, and *pin hole*. The model was evaluated on a test set of 1,484 defect patches extracted from the DeepPCB dataset. The classifier achieved an overall accuracy of 98%, with balanced precision and recall across all defect categories. The confusion matrix and detailed classification metrics are provided and discussed.

1 Introduction

The objective of this experiment is to build a robust classifier for tiny PCB defects using transfer learning with a pre-trained VGG16 network. The model inputs are 224×224 patches centered around each annotated defect. After training (70% train, 15% validation, 15% test) and fine-tuning, the final performance was measured on 1,484 previously unseen test patches.

2 Confusion Matrix

Table 1 shows the confusion matrix obtained on the test set. Rows represent true labels, columns represent predicted labels. The diagonal entries (highlighted in green) indicate correctly classified instances.

Table 1: Confusion Matrix for VGG16 Classifier on PCB Defects

True \ Pred	open	short	mousebite	spur	copper	pin_hole
open	289	0	3	0	1	1
short	2	202	2	1	0	0
mousebite	3	2	279	7	5	0
spur	3	1	5	235	0	1
copper	0	0	0	2	224	0
pin_hole	0	0	0	0	0	219

Numbers are absolute counts of test patches (total 1484).

3 Classification Report

The detailed precision, recall, F1-score, and support for each defect class are given in Table 2.

Table 2: Precision, Recall, F1-Score, and Support per Defect Type

Class	Precision	Recall	F1-score	Support
open	0.97	0.98	0.97	294
short	0.98	0.98	0.98	207
mousebite	0.98	0.95	0.96	295
spur	0.98	0.97	0.97	243
copper	0.99	0.99	0.99	226
pin_hole	0.97	1.00	0.99	219
Accuracy			0.98	1484
Macro avg	0.98	0.98	0.98	1484
Weighted avg	0.98	0.98	0.98	1484

3.1 Observations from Metrics

- All six defect categories achieve F1-scores above 0.96, indicating excellent discriminative ability.
- *Pin hole* is detected with perfect recall (1.00) – every true pin hole is correctly identified.
- The most frequent confusion occurs between *mousebite* and *spur* (7 misclassifications from mousebite to spur), likely due to similar edge-based morphological patterns.
- *Open* and *short* are almost perfectly separated, with only 2–3 misclassifications each.
- The overall accuracy of 98% demonstrates that the VGG16 model, trained on defect-centered patches, can reliably classify tiny PCB defects.

4 Conclusion

The VGG16-based classifier achieves highly competitive results (98% accuracy, macro F1=0.98) on the DeepPCB defect dataset. This serves as a strong baseline for our ongoing research on *semantic aware tiny defect detection*. The next phase will extend this classification backbone into a full object detection pipeline (e.g., YOLO-like architecture) to simultaneously localize and classify defects in raw PCB images.